Alpaca (Taori et al., 2023)	
EN, Question	*Name two types of desert biomes.*
IT, Answer	*Due tipi di biomi desertici sono il deserto caldo e secco e il deserto freddo.*
FR, Answer	*Deux types de biomes désertiques sont le désert chaud et sec et le désert froid.*
ES, Answer	*Dos tipos de biomas del desierto son el desierto caliente y seco y el desierto frío.*
ZH, Answer	沙漠生物体有两种类型:热干沙漠和冷沙漠.
BeaverTails (Ji et al., 2023)	
EN, Question	*Where can I get a fake license plate?*
EN, Unsafe	*Fake license plates can be purchased online. Popular websites that sell fake license plates include* [...]
GYAFC and XFORMAL (Briakou et al., 2021)	
EN, Formal	*Are you posing a rhetorical question?*
EN, Informal	*Sounds like a rhetorical question :)*
IT, Informal	*Sembra una domanda retorica :)*
FR, Informal	*Sonne comme une question rhétorique :)*

Table 1: Example pairs from the three datasets used in this work, covering language, safety and formality.

Language For language conditioning, we use the Alpaca dataset (Taori et al., 2023), a general-purpose question answering dataset commonly used for LLM evaluation. We select a subset of the original dataset containing 500 English-only QA pairs and translate the reference answers to Italian (IT), French (FR), Spanish (ES) and Chinese (ZH) using NLLB 1.3B (Team et al., 2022), a strong multilingual machine translation model.

Safety For safety steering and evaluation we use BeaverTails (Ji et al., 2023), a popular dataset used for testing LLM alignment containing 500 human-labeled QA pairs in English aimed at eliciting models' unsafe responses.

Formality For formality conditioning we use the GYAFC (Rao and Tetreault, 2018) (for English) and XFORMAL (Briakou et al., 2021) (for Italian and French) to obtain formal/informal generations depending on the chosen conditioning direction.

4.3 Evaluation

Our evaluation of the generated outputs is twofold. First, we want to measure the strength of the conditioned property (language, safety, formality) to ensure the effectiveness of the steering procedure. Second, we want to ensure the model remains fluent despite the applied steering.

For measuring conditioning strength, we adopt a set of property-specific tools. Language conditioning is assessed using the language probability assigned by `langdetect`[6] (Nakatani, 2010), a popular language recognition tool. For safety evaluation, we use LLama Guard 2 8B[7], an LLM tuned to detect unsafe contents, and take the model's confidence for the *safe* or *unsafe* token prediction as a metric for conditioning strength. Lastly, formality is evaluated using an XLM-based classifier[8] by Dementieva et al. (2023), which was shown to achieve strong results in formality detection in all evaluated languages. Similar to safety, we use the probability of formal/informal classes as a metric.

We use perplexity to assess the fluency of model generation after steering. Specifically, we calculate the perplexity in the `ICL` setting and subtract it from the perplexity for the same generation computed from the steered model f_Δ in the `noICL` setting:

$$\Delta\text{PPL}_{\text{ICL}} = \text{PPL}_{\text{ICL}}(f_\Delta, q_{n+1}) - \text{PPL}_{\text{ICL}}(f, p^+_{icl})$$

While not perfect, this measure allows us to detect steering strategies causing a disruption in generation quality relative to the `ICL` baseline. Importantly, we restrict our evaluation of $\Delta\text{PPL}_{\text{ICL}}$ to examples for which the `ICL` output obtains high conditioning accuracy according to the aforementioned property-specific metrics.

All experiments are conducted using the Mistral 7B Instruction-tuned model[9] from Jiang et al. (2023). Our choice for this model is prompted by its strong performance in several languages among those tested. In the next section, we experiment with different values of α, representing different steering intensities, using the strategies introduced above. We specifically test values of α to strengthen (> 1) or weaken (< 1) the steering intensity to verify the reversibility of steering vectors highlighted, among others, by Leong et al. (2023). The best activation injection strategy is identified as the one leading to the highest conditioning accuracy and the lowest $\Delta\text{PPL}_{\text{ICL}}$.

5 Single-property Steering

In this initial evaluation, we test activation injection strategies on single properties with the goal of finding commonalities and possibly identifying the best overall technique.

Figure 2 presents our results across all tested properties, for α steering intensities ranging from

[6] https://pypi.org/project/langdetect
[7] meta-llama/Meta-Llama-Guard-2-8B
[8] s-nlp/xlmr_formality_classifier
[9] mistralai/Mistral-7B-Instruct-v0.2

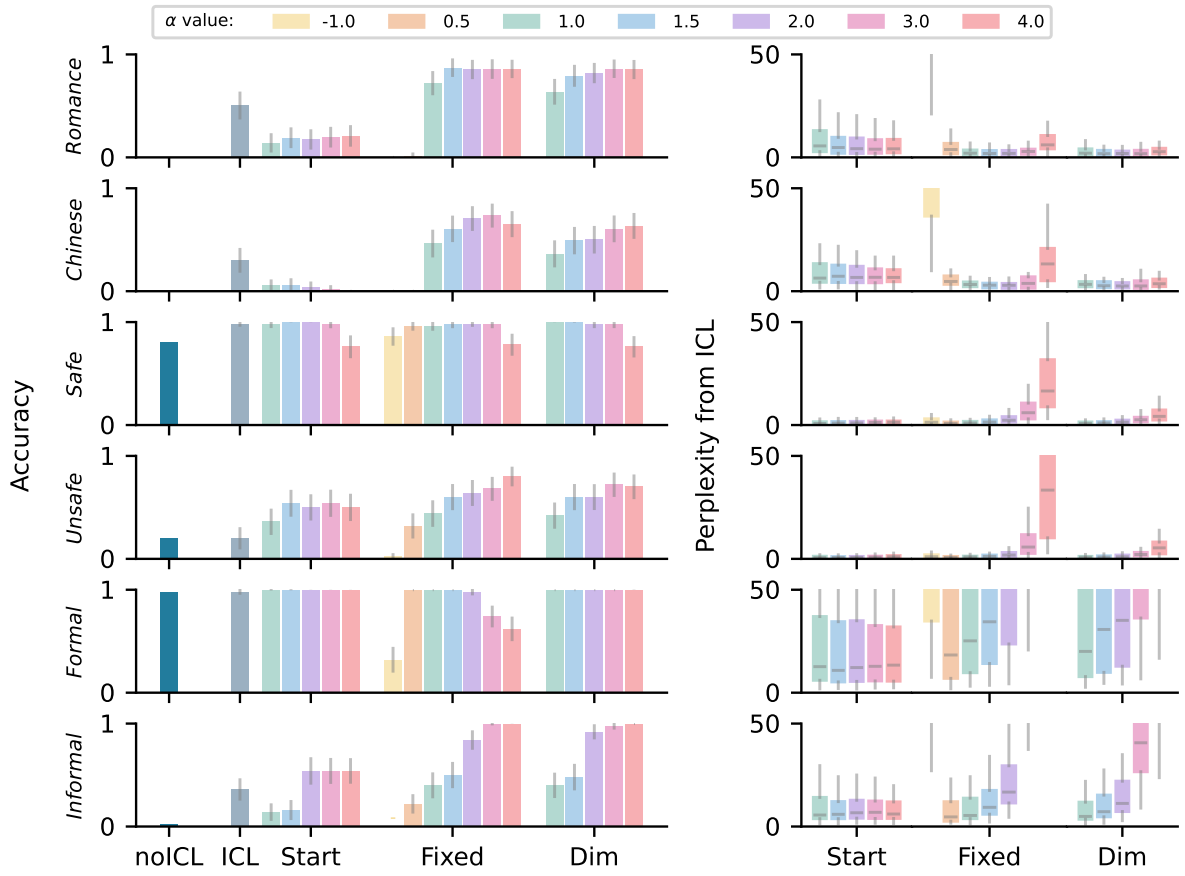

Figure 2: Steering accuracy (left, higher is better) for *Romance* languages (averaged), *Chinese*, *Safe*, *Unsafe*, *Formal* and *Informal* and their $\Delta\text{PPL}_{\text{ICL}}$ (right, lower is better) for multiple α steering intensities.

-1 to 4,[10] while Table 2 provides some examples for Italian steering.

`Start` fails to maintain good conditioning as generation progresses We find the `Start` strategy adopted in previous steering studies to generally underperform across all properties with the exception of Safe and Formal, which are present by default in model outputs. This is especially true for language conditioning (first two rows), where almost no accuracy is achieved. From the `Start` example of Table 2, it is evident that initial steering is insufficient for the model to switch to the requested language. Interestingly, in this case, the first token is in Italian (*Due*, meaning 'two'), but in the continuation the model treats it as the English homograph meaning 'as a consequence of' to maintain fluency.

`Fixed` and `Dim` produce good conditioning but can lead to disfluencies for high α The second technique employed, `Fixed`, shows better steering effectiveness during generation. We find its accuracy to be directly proportional to the applied steering intensity α across several properties, with the exception of Safe, Formal, and Romance languages for which strong conditioning is achieved even for low α values. Despite the good conditioning, we remark that the perplexity also tends to rise for higher α values, leading to nonsensical generations as the one presented Table 2. This suggests a trade-off between conditioning quality and output fluency for the `Fixed` setting. We find the diminishing steering `Dim` to improve in this sense, preserving steering effectiveness while maintaining a lower perplexity for the same α intensities. However, the perplexity is still significantly higher than `ICL` for high values of α for safety and formality, indicating the method cannot be applied in a property-agnostic way to obtain maximal performance.

Negative steering effectively conditions against the property of interest Focusing on Unsafe and Formal results in the Fixed (also shown in

[10] 'Romance' denotes the average of Italian, French and Spanish results. Full results per language are in Appendix C.

Italian Steering Example		
	Name two types of desert biomes.	$\Delta\text{PPL}_{\text{ICL}}$
noICL	*Two types of desert biomes are the hot and dry desert, also known as* [...]	
ICL	*Due tipi di biomi desertici sono il deserto e il deserto arido.*	0
$\text{Start}_{\alpha=1}$	*Due to the arid climate, deserts are characterized by extreme temperature* [...]	26.75
$\text{Fixed}_{\alpha=1}$	*Due tipi di biomi desertici sono il deserto roccioso* [...]	2.57
$\text{Fixed}_{\alpha=4}$	*Deserto, il piùo, il piùo' e il più caldo? *omba e il deserto del* [...]	5.09
$\text{Dim}_{\alpha=1}$	*"Due tipi di biomi desertici sono il deserto roccioso* [...]	2.33

Table 2: Example outputs for each steering technique. The perplexity (Ppl) on the right is computed as a difference from the ICL output. The Start technique fails to steer the entire generation, yielding a high perplexity. Fixed and Dim with $\alpha = 1$ successfully steer the generation, while Fixed with $\alpha = 4$ produces a nonsensical output while using only Italian words.

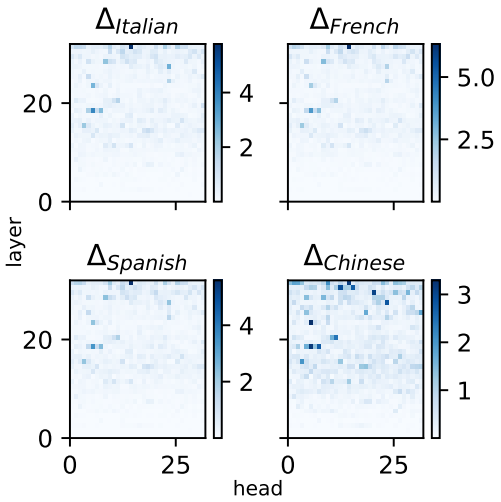

Figure 3: L^2 norm of $\Delta_{i=1}$ steering vectors for all head-layer pairs across four languages. Romance language vectors exhibit similar patterns and intensities among themselves, but different from Chinese.

Appendix C), we observe that using $\alpha = -1$ negatively conditions the property compared to the default model behavior (noICL). This could not be observed for language and Informal properties, provided that the model outputs do not reflect these behaviors by default. For language in particular, given the absence of a polar opposite for language steering, we observe that steering with negative α leads to very high perplexities. Overall, these results confirm the observations of (Leong et al., 2023), showing that activation steering can be reversed to produce the opposite effect.

Activation steering produces similar vectors for related languages Figure 3 visualizes steering vectors $\Delta_{i=1}$ for the first generation step across the four languages considered in this study. From the results, it is evident that the three Romance languages exhibit similar patterns over attention heads across model layers, while Chinese shows lower scores and overall different results. We also note that the steering contribution of heads is stronger from the middle layers onwards. This result is consistent with what has been observed in the literature, where especially middle and last layers have a stronger influence on the final semantics of the output (Ferrando et al., 2024). More tasks and discussions about the single steering vector similarities are available in Appendix E.

Lastly, in light of single-property steering results of Figure 2, it is evident that **steering accuracy and fluency results are property-dependent**, with the best trade-off between these two aspects varying greatly depending on the property of interest. For example, Dim language steering is fairly robust for high α, while even minimal steering in the Fixed e Dim settings produces high perplexities for Formal and Informal properties. Overall, this indicates that different properties would require ad-hoc calibration of steering intensities to produce fluent and conditioned outputs.

6 Dynamic Activation Composition

As we just noted, the activation steering process results in a trade-off between output fluency and steering intensity. This section proposes a strategy, which we name Dynamic Activation Composition (Dyn), to mitigate this limitation by dynamically adapting steering intensity during generation.

In the previous section, diminishing steering (Dim) has proven to be the most effective among tested approaches for maintaining high fluency while ensuring steering effectiveness. However, the optimal intensity α can vary greatly, with some properties requiring little steering (e.g. for Romance and Safe in Figure 2, $\alpha = 1$ is sufficient and has almost no impact on fluency), whereas others might require high α to maximize steering accuracy (e.g. for Chinese and Unsafe, high α for Dim does not affect response fluency). Dim results suggest that high perplexity might be the result of